

Supplementary Table S1

Bootstrap Stability Selection Frequencies for Student Stress Predictors.

Predictor	Selection_Frequency	Stability_Category
sleep_hours	1	Stable
study_hours	1	Stable
social_media_hours	1	Stable
attendance	1	Stable
exam_pressure	1	Stable
family_support	1	Stable
month7	1	Stable
month8	1	Stable
month9	1	Stable
month10	1	Stable
month11	1	Stable
month6	0.982	Stable
month3	0.962	Stable
month4	0.94	Stable
month2	0.904	Stable
month5	0.896	Stable
month12	0.838	Stable
student_typecollege	0.824	Stable
student_typeworking_student	0.71	Moderate
student_typeschool	0.654	Moderate

Supplementary Table S2

Random Forest SHAP Importance Values.

Feature	Mean_Absolute_SHAP
exam_pressure	0.179425651
study_hours	0.060311283
sleep_hours	0.033061522
family_support	0.031347736
social_media_hours	0.02540813
month	0.018957553
attendance	0.013183472
student_type	0.012225478

Supplementary Table S3.

Missing Data Summary.

variable	missing count	missing percent
student_type	0	0
sleep_hours	1333	5.23
study_hours	1277	5.01
social_media_hours	1312	5.15
attendance	1305	5.12
exam_pressure	1270	4.98
family_support	1291	5.06
month	1314	5.15
stress_level	0	0

Supplementary Figure S1.

Correlation Heatmap of Continuous Predictors.

